

Om Sanjaya Singhan

New York, NY

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EDUCATION

New York University

Sept 2023 - May 2025

MS, Computer Engineering

New York

- **Coursework:** Data Structures & Algorithms, Artificial Intelligence, Advanced Machine Learning, Deep Learning, Big Data, Network Security, Computer Systems Architecture, Real-time Embedded Systems, Machine Learning.

WORK EXPERIENCE

MAK Capital | Quantitative Software Developer | New York, NY, USA

Feb 2025 – May 2025

- Spearheaded a modular LLM analytics pipeline using OpenAI APIs, LoRA fine-tuning, and Python microservices, boosting alpha signal throughput by 40% with scalable performance.
- Developed a Python screening engine with SEC Edgar API, Pandas, and NumPy to flag insider sell-offs (50% holdings), using historical data pipelines and vectorized back testing for 20% downside capture on shorts.
- Integrated scalable ETL workflows with RESTful APIs (Yahoo Finance, SEC Edgar), leveraging async I/O and scheduled jobs to reduce false positives by 30% via market filters.

New York University | Research Assistant | New York, NY, USA

Sept 2024 – Present

NSF Project: Transforming Distribution System Situational Awareness via Continuous-Time Adaptive Data Fusion

- Led an AI simulation of power grid dynamics using TensorFlow and PyTorch, generating 150,000+ synthetic data points with 88% accuracy for real-time situational awareness.
- Enhanced a streaming pipeline with Node.js, Socket.io, and Redis caching, delivering grid data every 10 seconds to a React frontend, cutting latency by 45% with sub-500ms response times.
- Built D3.js dashboards with TimescaleDB and PostgreSQL, processing 75,000+ records with 50% improved retrieval efficiency, reducing outage prediction errors by 20% via predictive analytics.

Aroris Health | Software Developer | Minneapolis, MN, USA

May 2024 – Aug 2024

- Developed data encryption and validation scripts for API routes using Python and Firebase, ensuring HIPAA compliance and secure handling of 1000+ patient records.
- Collaborated and Optimized 10+ backend and frontend features with Node.js, React, GraphQL, and TypeORM, enhancing data integrity and user experience.
- Diagnosed and troubleshooted 20+ critical bugs by optimizing SQL queries and front-end components, resulting in a 30% performance improvement while reinforcing comprehensive testing methodologies.

PROJECTS

Personal AI Fitness Tool Using Retrieval Augmented Generation (RAG)

Oct 2024 – Dec 2024

- Built a fitness app with Streamlit and LangFlow workflows, generating personalized macronutrient and workout plans with real time user insights under 500ms response time.
- Integrated OpenAI models for query analysis, achieving 98% accuracy in tailored responses with API latency below 300ms.
- Managed 10,000+ user data points in Astra DB, leveraging vector search for contextual retrieval with 95% query accuracy for AI responses.

Music Recommendation System

Dec 2024 - Feb 2025

- Engineered a music recommendation system using Spotify/Genius APIs with Streamlit, leveraging Spotipy and OAuth for Spotify authentication, achieving 85% accuracy for 500+ song playlists via scikit-learn TF-IDF and cosine similarity.
- Crafted an NLTK-based NLP pipeline to process 10,000+ lyrics, filtering non-English text with langdetect, cutting preprocessing latency by 40% using optimized pandas operations.
- Enabled async Spotify/Genius API calls, caching 1,000+ responses to reduce response time by 35%, with D3.js hover effects for song posters.

Enhanced Resume/CV Matching Using Doc2Vec and Deep Learning

Jan 2024 - May 2024

- Automated a semantic job-match system with Doc2Vec, achieving 85% accuracy in aligning resumes to job descriptions using Kaggle and Hugging Face datasets for the tech industry.
- Applied NLP techniques with NLTK, SpaCy, tokenization, and lemmatization, using Gensim's Doc2Vec for embeddings and cosine similarity for precise ranking.

SKILLS & CERTIFICATIONS

- **Programming Languages:** Python, Java, JavaScript, TypeScript, C++/C, C#, HTML/CSS.
- **Web Technologies & Database:** React.js, Node.js, GraphQL, Redis, Express, SQL, MongoDB, Postgres, TypeORM, Django.
- **Machine Learning & DevOps:** Matlab, Pytorch, Sci-kit Learn, TensorFlow, Numpy, Pandas, OpenAI, LangFlow, Langchain, Azure, AWS, Terraform, Kubernetes, Docker, CI/CD.
- **Development Tools & Communication:** Git, Jira, Slack, Excel, Linux.
- **Certifications:** IBM certified Applied AI Professional, Docker Professional Certificate.